Course Code: MTH3705	**Course Name:** Applied Machine Learning in Health Care	
Credits: 3-0-0	**Contact hours per week:** 3 lectures per week	
Batch: 2023, 6th Sem **Academic Year:** 2025-26	**Semester Duration:** 19th Jan - 05th May 2026	
Instructor-in-charge: Dr Palak Goel Email – palak.goel@bmu.edu.in Phone -7838887146 Office –23, Fourth Floor, E2 Building	**Course Coordinator –** Dr Palak Goel Email – palak.goel@bmu.edu.in Phone -7838887146 Office –23, Fourth Floor, E2 Building	

Aim of the course: In the post-COVID-19 era, understanding disease dynamics, analyzing epidemiological data, and building predictive models have become crucial skills for engineers working in healthcare technology, data science, public health informatics, and AI-driven health solutions. This course is designed to equip B.Tech. students with practical skills in epidemic modeling, data analysis, and visualization using Python.

Course Overview:

This is an introductory undergraduate course on epidemic modeling and infectious disease data analysis using Python. The focus will be on epidemic modeling and infectious disease data analysis using Python, with an emphasis on combining theoretical understanding and hands-on programming. The course covers the analysis and visualization of real-world epidemiological datasets, development of compartmental disease models, and application of statistical and machine learning techniques for prediction and forecasting. Salient topics of the course are listed below.

Week-Wise Schedule:

Note: *Weekly course schedule includes 03 lecture sessions.*

Week 1-3: Introduction to Public Health Data, Epidemiological Data Wrangling, and Analysis
Understanding "Line lists," incidence vs. prevalence, and the "Chain of Infection", Fetching live data, Handling biological data, Pivot tables for time-series structure, Moving averages to smooth daily infection spikes. Incidence mapping and choropleth maps.

Week 4-7: The Physics of Pandemics (Compartmental Models) and Parameter Estimation
Introduction to epidemic modeling (mechanistic vs phenomenological), Compartmental Models, Parameter estimation: least squares and maximum likelihood, Basic reproduction number (R0) and herd immunity, Effective reproduction number (Rt), Model validation, Implementing and simulating compartmental models in Python.

Week 8-10: **Time Series & Forecasting**
Introduction to time series analysis, Time series objects and data structures, Stationarity and differencing, Trend estimation and smoothing, Seasonal decomposition (additive and multiplicative), Autocorrelation and partial autocorrelation, Simple forecasting methods (naive, moving average, drift), ARIMA models, Model selection, Cross-validation for time series, Residual diagnostics and forecast accuracy metrics

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Week 11-12: **Regression and Classification Methods**
Linear Regression, Multiple Linear Regression, Polynomial regression, Logistic Regression, Poisson regression using public health data.

Week 13-16: **Ensemble Modelling**
Introduction to neural networks; Architecture: neurons, layers, activation functions; Forward propagation and backpropagation; Overfitting and regularization techniques; Applications to epidemic forecasting.

Course Outcomes:

At the end of the course, the students should be able to:

Course Outcome Statement	Mapped Component	Associated Activities / Assessment
CO1: Understand the fundamentals of public health data and epidemiological concepts including line lists, incidence and prevalence measures, chain of infection, biological data handling, time-series structuring, smoothing techniques, and spatial visualization using incidence maps and choropleth maps.	Skill Development	Quiz, Assignment, Mid-Term Presentation & End-Term Presentation for project work
CO2: Apply data wrangling, visualization, and computational techniques for real-world epidemiological datasets, fetch live public health data, construct time-series datasets, and implement compartmental epidemic models with parameter estimation and reproduction number analysis.	Skill Development	Quiz, Assignment Mid-Term Presentation & End-Term Presentation for project work
CO3: Analyze epidemic dynamics using compartmental models, time-series analysis, forecasting techniques, regression models, and evaluate	Skill Development Employability	Assignment Mid-Term Presentation & End-Term Presentation

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model performance using validation metrics, residual diagnostics, and comparative analysis between classical and data-driven models.		for project work
CO4: Evaluate and formulate real-world infectious disease problems by integrating epidemiological theory, statistical modeling, time-series forecasting, and neural networks to develop predictive dashboards and decision-support tools for public health applications.	Skill Development Employability	End-Term Presentation for project work

CO-PO Mapping

PO and PSO →	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**	**PSO3**	**PSO4**
↓ CO Mapping || Course Mapping →																
CO1	1			1			1						1			
CO2	3	2		2	2		2						2			
CO3	3	3	2	2	3	3	3						3	3		
CO4	3	3	3	2	3	3	3		2	2		2	3	3		

For details about PO, PSO, and CO-PO mapping, refer to OBE document for B. Tech. – CSE program.

1- Low Correlation; 2- Moderate Correlation; 3- Substantial Correlation

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Course Competencies (Course Outcomes further elaborated) and Instruction schedule:

	Competency	CO	CL	No of Contact hours
C1	**Introduction to Public Health & Epidemiological Data** - Public health data sources - Line lists and surveillance data - Incidence vs. prevalence - Chain of infection - Biological and epidemiological data handling	CO1	U AP	6
C2	**Epidemiological Data Wrangling & Visualization** - Fetching live epidemiological data - Data cleaning and preprocessing - Pivot tables and time-series restructuring - Moving averages and smoothing - Incidence mapping and choropleth maps	CO1, CO2, CO4	U AP EV	6
C3	**Physics of Pandemics & Compartmental Models** - Mechanistic vs phenomenological models - SIR, SEIR and extensions - Model assumptions and limitations - Numerical simulation of compartmental models.	CO1, CO2, CO3, CO4	U AP AN EV	9
C4	**Parameter Estimation & Reproduction Numbers** - Least squares estimation - Maximum likelihood estimation - Basic reproduction number (R_0) - Effective reproduction number (R_t) - Herd immunity threshold and model validation	CO1, CO2, CO3, CO4	U AP AN EV	9
C5	**Time Series Analysis in Epidemiology** - Time series concepts and data structures - Stationarity and differencing - Trend estimation and smoothing - Seasonal decomposition	CO1, CO2, CO4	U AP	6
C6	**Forecasting Models for Disease Spread** - Autocorrelation and partial autocorrelation - Naive, moving average and drift methods - ARIMA models - Model selection and cross-validation - Residual diagnostics and forecast accuracy metrics	CO1, CO2, CO3, CO4	U AP AN EV	6
C7	**Regression and Neural Networks for Public Health** - Linear, logistic and Poisson regression - Neural network architecture and training - Overfitting and regularization - Epidemic forecasting using neural networks - Model development and final project	CO1, CO2, CO3, CO4	U AP AN EV	6

Total contact hours=48

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Learning Resources:

Textbooks:

1. Ding, Peng. Epidemic Modeling and Analysis with R. CRC Press, 2023. (Python adaptations will be provided for all R code examples).

Reference Books:

1. McKinney, Wes. Python for Data Analysis: Data Wrangling with Pandas, NumPy, and Jupyter. 3rd Edition, O'Reilly Media, 2022.
2. VanderPlas, Jake. Python Data Science Handbook: Essential Tools for Working with Data. O'Reilly Media, 2016.
3. Keeling, Matt J., and Pejman Rohani. Modeling Infectious Diseases in Humans and Animals. Princeton University Press, 2008.
4. Vynnycky, Emilia, and Richard G. White. An Introduction to Infectious Disease Modelling. Oxford University Press, 2010.

E-Learning (and other) Resources:

1. https://www.statsmodels.org/stable/tsa.html
2. https://people.duke.edu/~rnau/411arim.htm
3. https://www.youtube.com/c/statquest
4. https://www.who.int/data
5. https://www.kaggle.com/datasets?search=epidemiology

Assessment Pattern:

The final grade will be based on the marks obtained in various assessment components defined in the assessment table below. Relative grading method defined in the academic regulations of the university will be followed to grade the students.

Evaluation Component	Weightage (%)	Evaluation Schedule	Rubrics/Remarks
Quiz	20 (2 quizzes each of weightage 10%)	February 23, 2026 April 16, 2026	Each quiz is a 20-minute-long closed-book assessment component. Understanding of Concepts (40%) ● Epidemiological concepts such as incidence, prevalence, compartmental models, and time-series basics. ● Validation of students' understanding through timed quiz. Problem Solving (60%) ● Application of data analysis, modeling, and forecasting techniques.

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			● Logical ability to problem-solving skill in a timely manner.
Assignment	20 (1 assignment of weightage 20%)	April 10, 2026	One open-book assignment of weight 10%. Understanding of Concepts (20%) ● Detailed understanding of epidemiological data structures, modeling assumptions, and forecasting methods. ● Ability to interpret outputs from computational tools. Timeliness (20%) ● Assignment must be submitted on or before the due date and time. Problem Solving (60%) ● Application of computational and statistical methods to real-world public health datasets. ● Interpretation and discussion of analytical results.
Mid-Term Presentation (Equivalent to Mid-Term Examination)	20	March 29, 2026 (Presentation tentatively between March 30-April 03 2026)	Understanding of Concepts (20%) ● Clarity in explaining epidemiological concepts, data sources, and modeling approach. ● Ability to relate theory with real-world disease data. Timeliness (10%) ● Presentation submission on or before the specified deadline. Problem Solving (20%) ● Demonstration of coding skills. ● Visualization of the problems using software. Presentation and Viva (50%) ● Logical structure, clarity of slides, and effective communication. ● Developing presentation skills to present the work in a group. ● Demonstration of individual knowledge through viva.
Project	40	April 22 2026	Understanding of Concepts (10%)

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(Equivalent to End-Term Examination)		(Presentation tentative April 23 - May 1 2026)	● Comprehensive understanding of epidemiological theory and computational techniques. Timeliness (10%) ● Submission of project report and code within the stipulated deadline. Report Writing and Problem Solving (30%) ● Preparation of a structured technical report. ● Application of modeling, forecasting, and data analysis methods to a real disease dataset. ● Demonstration of programming skills and result interpretation. Presentation and Viva (50%) ● Clear presentation of methodology, results, and conclusions. ● Demonstration of individual contribution and understanding through viva. ● Ability to apply concepts to solve complex problems

Experiential Learning Components:

Hands-on projects on applications of Computational Epidemiology and Public Health Data Analytics using Python will be given to the students. Students will work in groups on real-world public health datasets involving epidemiological data wrangling, implementation of compartmental epidemic models, parameter estimation, time-series forecasting, and visualization of disease trends through dashboards. This entire set of experiential learning activities will be assessed through the evaluation component carrying a weightage of 60%.

Student Responsibilities:

- Attend lectures to obtain all the course material that you are responsible for.
- Check announcements at Maitri/Google Classroom on a regular basis.
- Submit assignments/projects on time.

Regularly, check your marks on the Maitri and inform the instructor in-charge in case of any issues.

Attendance Policy: Students are expected to attend the classes regularly. Failure to attend the classes regularly and adhere to the expected attendance percentage may result in a reduction of the grade or awarding 'R' grade as per the University's grading policy.

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Recourse examination policy: As per institute's policy.

Late assignment submission policy: Penalty of 1 mark for each day of delay in submitting the assignment.

Make-up examination/work policy: As per University policies.

Behavior expectations: No mobile phones and other distractive elements and gadgets in the class.

Other Policies or Procedures:

a) Academic Honesty

Students must abide by BMU's academic honesty policy. For example, an excerpt is provided for guidance as follows:

"Students who compromise the integrity of the classroom are subject to disciplinary action on the part of the University. Violations of classroom standard include:

1. Cheating in any form, whether in formal examinations or elsewhere
2. Plagiarism, using the work of others as your own without assigning proper credit to the source.
3. Misrepresentation of any work done in the classroom or in preparation for class.
4. Disruptive behavior in a course of study or abusiveness toward faculty or fellow students."

b) Teaching Improvement

Please feel free to make suggestions to improve the content of the class and my instruction skills. You can share these suggestions directly to me and/or via email. You can also leave your comments anonymously by sending online messages.

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